CONVOLUTIONAL NEURAL NETWORKS FOR BIOMEDICAL IMAGE SEGMENTATION

Group 136: Anton Wang Strandberg (s183220), Ona Šaulianskaitė (s232958) & Natasha Garibay (s134735)

1. INTRODUCTION

1.1. Image segmentation and U-net

Image segmentation is the process of dividing an image into multiple segments or regions, which simplifies the analysis and extraction of meaningful information from raw pixel data. By segmenting an image, we assign context to individual pixels, transforming raw visual data into structured information that machines can interpret. Among various segmentation techniques, U-Net has emerged as a state-of-the-art model, particularly advancing biomedical image analysis (Figure 1).

U-Net's distinctive U-shaped architecture consists of two main components: the encoder (contracting path) and the decoder (expanding path). The encoder takes the input image, downsamples its spatial dimensions through convolutional and max-pooling layers, and extracts high-level features. The decoder upsamples these feature maps using techniques like transposed convolutions, reconstructing the image with the original spatial dimensions. A key feature of U-Net is the use of skip connections, which directly transfer low-level features from the encoder to the decoder, helping maintain fine spatial details [1].

1.2. Cell segmentation using Cellpose

Cell segmentation plays a crucial role in microscopy image analysis, allowing for precise identification of cellular structures, cell counting, and analysis of cell morphology. Cellpose is a deep learning-based algorithm designed for automatic cell segmentation in microscopy images. It can be fine-tuned for custom datasets and requires minimal annotated data for training [2].

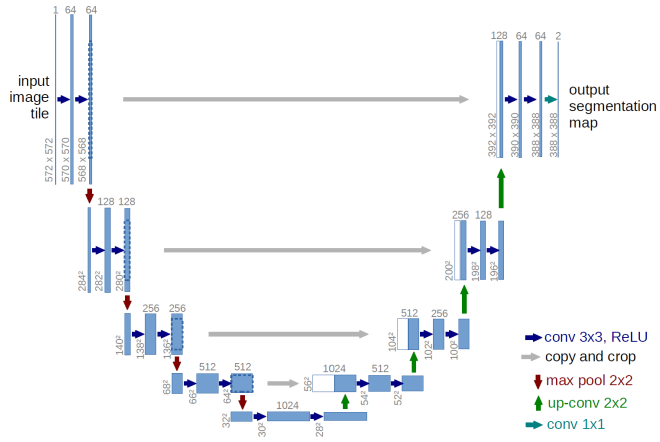


Fig. 1: This image illustrates the U-Net architecture for image segmentation, featuring an encoder-decoder structure. Input tiles are down-sampled via convolutions and max-pooling (red arrows) and up-sampled with transposed convolutions (green arrows). Skip connections (gray arrows) retain spatial details, leading to the final segmentation map after a 1x1 convolution. Feature map dimensions and channel counts are annotated.

To train a Cellpose model, annotated data (raw images and predicted masks) are provided as input (Figure 2). Simulated diffusion is applied to the ground-truth masks to create topological maps, which represent intensity gradients. These maps have higher intensity values at the center of the cell and lower values at the boundaries. The neural network is then trained using these maps to learn the spatial distribution of cell structures. The network predicts both vertical and horizontal gradients, indicating how pixel intensities change in these directions.

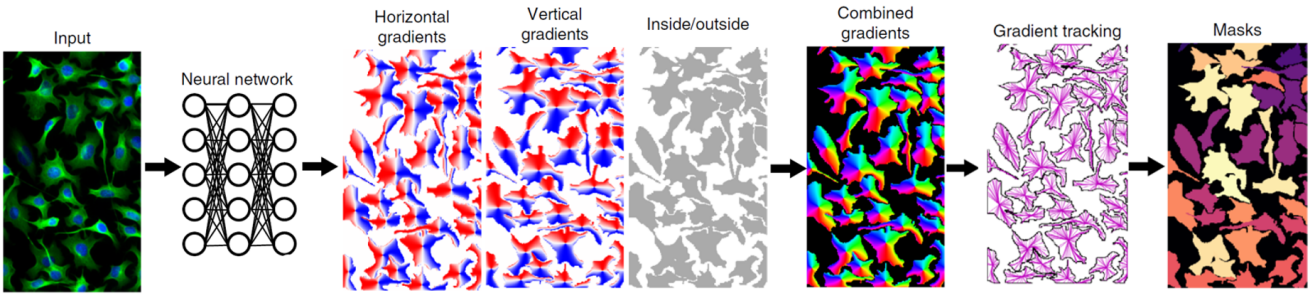


Fig. 2: The image shows the Cellpose model workflow. Annotated input images are processed by the neural network to predict horizontal and vertical gradients and a binary mask (inside/outside cell). These gradients are combined into vector fields, enabling gradient tracking where pixels converge to shared points. Pixels converging to the same point are grouped into a single mask.

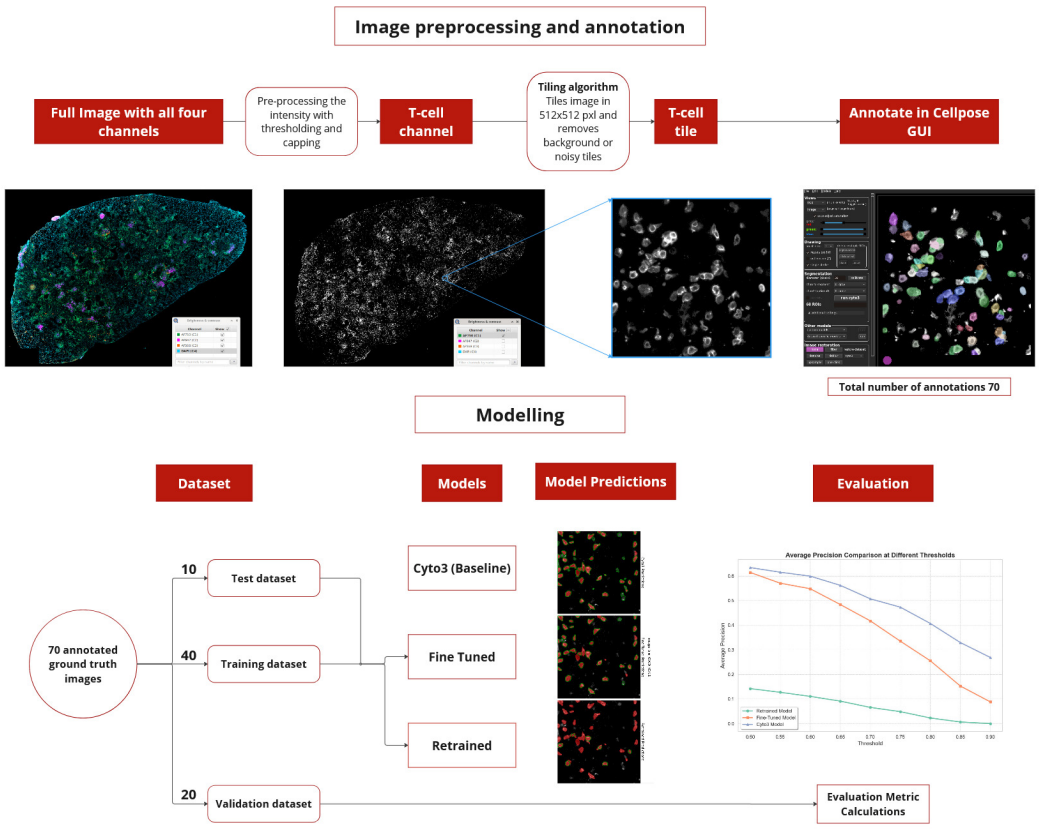


Fig. 3: Overview of our workflow in this project. Created using Miro[3]

It also predicts a binary map that classifies each pixel as either being inside or outside the cell. By combining these gradients, a vector field is formed, where each pixel has a vector showing the direction and strength of intensity change. Gradient tracking follows these vectors, guiding the model to segment cells by grouping pixels that converge to the same point, allowing for precise cell segmentation and shape recovery.

1.3. Project Aim

The goal of this project was to evaluate the performance of the Cellpose model by comparing its baseline functionality with results obtained through retraining and fine-tuning using our annotated data, in order to assess the impact of these approaches on segmentation accuracy.

2. MATERIALS AND METHODS

Figure 3 provides an overview of the project's workflow, detailing the process from training the image dataset to evaluating the three different models. The code can be found at: Cell Segmentation DL Project. All implementations were carried out in Python 3 [4] executed through the DTU HPC terminal [5].

2.1. Data

We used a dataset provided by Statens Serum Institute, from a vaccine study on *Mycobacterium Tuberculosis*. The dataset includes 32 images of infected mouse lungs, collected at 2, 4, 6, and 16 weeks. The mice were divided into two groups: a control group (unvaccinated) and a vaccinated group.

Each image represents a cross-section of lung tissue, with cells stained using four fluorescent markers: CD4+ (T-cells), B202 (B cells), MECA (HEV), and DAPI (nuclei). For this project, we focused solely on the T-cell stained cells. The images are approximately 400,000 pixels (20,000 x 25,000 pixels), making annotation time-consuming. As a result, only one image was used for annotation and analysis, though we expect the model to perform well on the other images as well.

To annotate the image, we employed a tiling algorithm that divided the image into 512×512 pixel tiles. Initially, we encountered significant number of static noise and background tiles. To address this issue, we employed a filtering strategy based on the tile's average intensity and overall variance. If the variance of a tile was below 10, it was classified as containing only background or 0-intensity pixels. Additionally, tiles with variance below a certain noise threshold were flagged as static. In total, the image generated 841 tiles for

analysis in the T-cell channel.

2.2. Annotation in Cellpose GUI

Out of approximately 850 tiles in the T-cell channel, 70 tiles were randomly selected for annotation. Annotations were generated using the Cellpose GUI [6] with the cyto3 model [7], where the segmentation channel was set to 0 and the optional channel (typically used for nuclei) was left as `None`. After the initial automatic annotation, we manually refined the results by adding annotations where Cellpose had missed cells, deleting annotations that did not represent cells, and splitting any annotations that merged two or more cells.

The newly annotated images were saved to be used for training (40 images), testing (10 images), and validation (20 images) for three different models: the baseline cyto3 model, a fine-tuned model, and a retrained model.

2.3. Model training

The **fine-tuned model** was trained using the Cyto3 model, which had already been pretrained on a large-scale dataset for cell segmentation tasks. This approach leverages transfer learning, where the knowledge acquired by the Cyto3 model is applied to our specific dataset to improve performance with limited annotated data. The pretrained weights from the Cyto3 model provided a strong initialization, allowing the model to focus on refining features relevant to our dataset.

We retained a relatively high learning rate of 0.1, consistent with the recommendations in the original paper, and applied a weight decay of 0.00001 to mitigate overfitting. The model was trained for 100 epochs using stochastic gradient descent (SGD) with a batch size of 8 (default parameter). Training was performed on 40 images, with 10 images reserved for testing to evaluate the model's generalization.

The **retrained model** used similar hyperparameters to ensure a consistent training approach. However, unlike the fine-tuned model, the retrained model was initialized from scratch, without utilizing pretrained weights from the Cyto3 model. This allowed the model to learn entirely from our dataset using the standard Cellpose U-Net architecture, with all layers initialized randomly. For training, 40 images were used, with 10 images reserved for testing to evaluate the model's generalization.

2.4. Model evaluation

For evaluation we chose the average precision (AP) metric, which was also used to evaluate the original cellpose model performance [2]. AP is defined as:

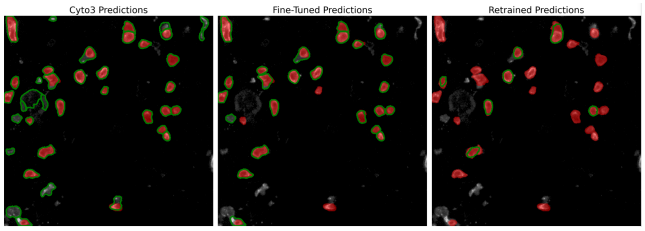
$$AP = \frac{TP}{TP + FP + FN}$$

TP = True positives, FP = False positives and FN = False negatives.

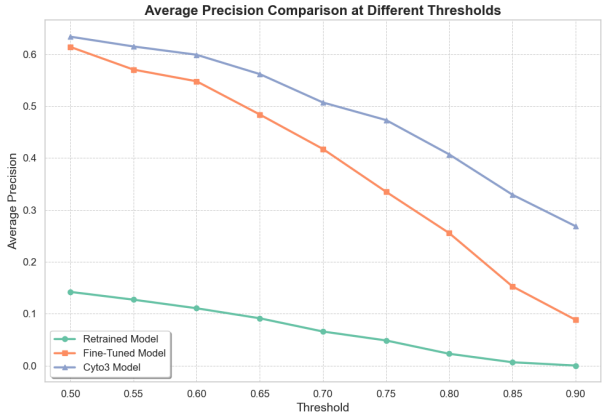
Cellpose employs AP to assess segmentation performance by measuring object-level accuracy. In this method, true positives are determined based on the Intersection-over-Union (IoU) between predicted and ground truth cell masks. A prediction is considered a correct detection if its IoU exceeds a threshold of 0.5. The results are presented in Figure 4b, highlighting the performance differences among the three models. We also evaluated the models' performance visually using one of the validation images (Figure 4a). Additional images are available on the GitHub results section .

3. RESULTS & DISCUSSION

The results showed that the original model, Cyto3, had the highest precision and outperformed both the fine-tuned and retrained models. The second best model was the fine-tuned one, where we trained the model using the Cyto3 model as a base (pretrained model) and used our annotated images.



(a) Segmentation vs groundtruth annotations.
1. **Consensus (True Positives):** cells that are both colored in red (ground truth) and outlined in green (model predictions).
2. **False Negatives:** cells that are colored in red but not outlined in green.
3. **False Positives:** objects that are outlined in green but not colored in red.



(b) Average precision of the three different models

Fig. 4: The resulting three models performance was assessed by: a) visually inspecting the predicted segmentation vs the groundtruth segmentation. And b) by comparing the average precision over a range of thresholds for overlapping predictions and groundtruths

The model that performed the worst was the retrained model, which we trained using our annotated images (Figure 4b). First, using the Cyto3 model for annotating images may have introduced some bias favoring the two best-performing models, Cyto3 and the fine-tuned model. Additionally, both Cyto3 and the fine-tuned models were trained on a large dataset, therefore, benefited from extensive training. In contrast, our retrained model was developed using only 40 annotated images, significantly limiting its exposure to diverse training data.

We initially anticipated that the fine-tuned model would outperform the original Cyto3 model, as it leveraged both the original training and additional training on our 40 annotated supplementary images. However, this was not the case. A likely explanation is the inconsistency in our annotations, stemming from our initial unfamiliarity with the images. These inconsistencies may have limited the fine-tuned model's ability to effectively learn new patterns from our data, ultimately hindering its performance advantage over the original Cyto3 model.

These results can also be visually confirmed (Figure 4a and the other images that are available on the GitHub results section). Although Cyto3 is the best-performing model, with the majority of its predictions overlapping the ground truth masks, it has a tendency to overpredict. In contrast, the fine-tuned model, while also performing fairly well with substantial overlap between the ground-truth and predicted masks, is more conservative and tends to underpredict. This means it generates fewer false positives compared to Cyto3 but at the cost of missing true positives. The retrained model clearly underperforms, with minimal overlap between the ground-truth and predicted masks. It is highly conservative, resulting in a significant number of false negatives. However, this conservatism also eliminates false positives, as the model avoids overprediction entirely.

One possible reason for the fine-tuned model underperforming and being more conservative is that we used the Cyto3 model as a baseline, inheriting its initial weights and parameters, and trained it solely on our 40 annotated images. While this approach allowed the model to adapt to our dataset, it may have limited its ability to generalize to broader patterns.

The fine-tuned model could have benefited from training on a combined dataset, including both the original Cellpose dataset and our annotated images. This would have provided foundational features from the original dataset, enhancing the model's ability to recognize shared cell structures and better generalize to our data. A combined approach might have improved the model's balance between true positive detection and false positive reduction, leading to better overall performance.

Due to the limited time frame of this project, several potential strategies and tweaks to improve performance could not be explored. For instance, we did not experiment with different sets of hyperparameters, such as the choice of optimizer, number of epochs, normalization methods, weight decay, and learning rate. The model, in particular, could have benefited from a smaller learning rate, as this would help prevent overfitting by allowing the parameters to converge more gradually and avoid prematurely settling at local minima.

Regarding the retrained model, we lacked the time and resources to acquire a sufficiently large dataset of annotated images. In theory, this model could have achieved the best performance if trained on a dataset comparable in size to the one originally used to train the Cellpose model (8,000 images). A retrained model with this scale of data would likely have been highly specialized for segmenting images specific to our project.

Our dataset included only a single channel for T-cell staining, which limited both the annotation process and the models' performance. First of all, incorporating a second channel, such as a nuclei channel, would have significantly improved our ability to annotate the images. This additional channel would have allowed us to better distinguish individual cells in densely packed regions with no clear separations. Moreover for predictions, the second channel could have helped the models differentiate cells from light artifacts in the images. For instance, the overlap of T-cell staining with a nuclei signal would indicate a true cell, whereas T-cell staining without a nuclei signal would likely indicate an artifact. This improvement would have been particularly impactful for the retrained model, as it was trained exclusively on our images and relied solely on the T-cell channel. Incorporating the nuclei channel, could have helped the model learn spatial patterns more effectively — for example, using the nuclei channel to localize cell centers and the T-cell channel to define cell boundaries. This might have reduced false negatives and improved segmentation in densely packed regions. Similarly, the fine-tuned model would have benefited from more accurate annotations enabled by the nuclei channel, enhancing its overall segmentation performance and robustness.

In conclusion, our results highlight the importance of dataset size and model generalization. The Cyto3 model outperformed both the fine-tuned and retrained models, with the fine-tuned model showing conservative behavior but generating fewer false positives. The retrained model, limited to 40 annotated images, underperformed due to insufficient data diversity. Combining the original Cellpose dataset with our data could have improved generalization and overall performance. Future improvements, such as optimizing hyperparameters, increasing annotated dataset size, and incorporating additional channels like nuclei, could further enhance model accuracy and robustness.

4. REFERENCES

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